

Machine Learning B (2025)

Home Assignment 7

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1 XGBoost

Let a dataset $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\} \subset \mathbb{R}^{10} \times \mathbb{R}$, which is stored in the file `quasars.csv`. Here, each line contains, for each object (row), a target value $y_i \in \mathbb{R}$ (last column) and a vector $\mathbf{x}_i \in \mathbb{R}^{10}$ of ten attributes (first 10 columns). In the following, we will build XGBoost regression models to predict the target (redshift) for new, unseen objects.

Response to Question 1. First, we load `quasars.csv` into a Pandas DataFrame, and separate the features (first 10 columns) from the target value (last column), denoting these X and y , respectively. Then, we split the data into a training set containing 80% and a test set containing 20% of the instances by using `train_test_split` from `scikit-learn`:

```
df = pd.read_csv("quasars.csv", header=None)

X = df.iloc[:, 0:10].values
y = df.iloc[:, 10].values

X_train_full, X_test, y_train_full, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)
```

Listing 1: Code to load the dataset and split it into training/test sets.

Response to Question 2. Next, we will fit a XGBoost regression model on about 90% of the training data and make use of an hold-out validation set containing about 10% of the training set to monitor the training process with RMSE curves. The final 20% (`X_test`, `y_test`) remains untouched until the very end. We build the XGBoost Regressor by using the `XGBRegressor` class from the `xgboost` library in Python, where we use the following parameter assignments (default assignments are used for the remaining ones):

Parameter	<code>colsample_bytree</code>	<code>learning_rate</code>	<code>max_depth</code>	<code>reg_lambda</code>	<code>n_estimators</code>
Value	0.5	0.1	4	1	500

Table 1: Parameters used for fitting the XGBoost regression model.

For this regressor, we set the `objective` parameter to `"reg:squarederror"`, which is regression with squared loss. This setting ensures that the model minimizes the squared error between the predicted and true values. Also, we want to record, at each boosting round, both training-set RMSE and validation-set RMSE. `xgboost`'s `.fit(...)` method can accept `eval_set=[(X_train, y_train), (X_val, y_val)]` and `eval_metric="rmse"`, so that it reports (and records) RMSE on both sets at every iteration. The code is seen below:

```
X_train, X_val, y_train, y_val = train_test_split(
    X_train_full, y_train_full, test_size=0.10, random_state=1
)

xgb_base = xgb.XGBRegressor(
```

```

    objective="reg:squarederror",
    colsample_bytree=0.5,
    learning_rate=0.1,
    max_depth=4,
    reg_lambda=1,
    n_estimators=500,
    random_state=42,
    eval_metric="rmse"
)

xgb_base.fit(
    X_train, y_train,
    eval_set=[(X_train, y_train), (X_val, y_val)],
    verbose=False
)
evals_result = xgb_base.evals_result()

```

Listing 2: Code to fit an XGBoost with prescribed hyperparameters + 10% hold-out from training.

Then, by extracting both `evals_result['validation_0']['rmse']` (training set) and `evals_result['validation_1']['rmse']` (validation set), we have plotted RMSE on the y-axis against iteration number on the x-axis with `matplotlib`. Figure 1 below shows these RMSE curves for both the training and validation sets as a function of the boosting iteration. As expected, we see that the training RMSE decreases steadily, while the validation RMSE plateaus, indicating the model's generalization performance.

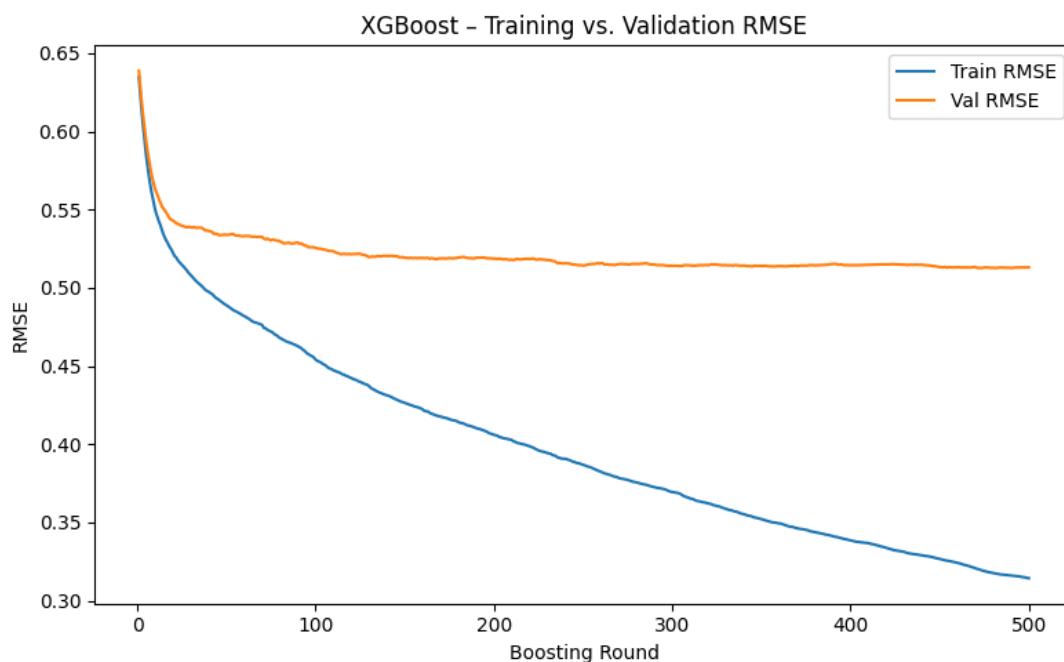


Figure 1: RMSE curves for both training and validation sets as a function of boosting rounds.

After training, we make predictions via the fitted model on the held-out test set (20% of the original data) by `y_pred_test = xgb_base.predict(X_test)`. We report the RMSE by

`np.sqrt(mean_squared_error(y_test, y_pred_test))` and coefficient of determination (R^2) by `r2_score(y_test, y_pred_test)` to assess predictive accuracy on unseen data:

- Test RMSE: 0.4989
- Test R^2 : 0.3214

Response to Question 3. We will now conduct a grid search to find good parameter assignments for our XGBoost model. Here, we must include the parameters used above in Table 1 as a starting point and then extend the grid with additional values for `colsample_bytree`, `learning_rate`, `max_depth`, `reg_lambda`, and `n_estimators`, each varied around the values used in the initial model. The hyperparameter grid is seen below:

```
param_grid = {
    # vary colsample_bytree around 0.5
    "colsample_bytree": [0.3, 0.4, 0.5, 0.6, 0.7],
    # vary learning_rate around 0.1
    "learning_rate": [0.01, 0.05, 0.1, 0.2, 0.3],
    # vary max_depth around 4
    "max_depth": [2, 3, 4, 5, 6],
    # vary reg_lambda around 1
    "reg_lambda": [0.1, 0.5, 1.0, 2.0, 5.0],
    # consider fewer / more boosting rounds
    "n_estimators": [200, 300, 500, 700, 900]
}
```

Listing 3: Grid of selected hyperparameters for grid search with 3-fold CV.

This grid has $5 \times 5 \times 5 \times 5 \times 5 = 3125$ total combinations, and since we will fit 3 folds for each of the 3125 candidates, it will total in $3 \times 3125 = 9375$ fits. The grid search is implemented using `GridSearchCV` from `scikit-learn`, with the negative root mean squared error (`neg_root_mean_squared_error`) as the scoring metric. The grid search evaluates all combinations of the specified hyperparameters, training and validating the model using 3-fold cross-validation for each combination. The set of hyperparameters that yields the lowest average validation RMSE is selected as the best configuration:

```
xgb_for_gs = xgb.XGBRegressor(
    objective="reg:squarederror",
    random_state=42
)

grid_search = GridSearchCV(
    estimator=xgb_for_gs,
    param_grid=param_grid,
    scoring="neg_root_mean_squared_error",
    cv=3,
    verbose=1,
    n_jobs=-1
)

grid_search.fit(X_train_full, y_train_full)
best_params = grid_search.best_params_
print(best_params)
```

```
Fitting 3 folds for each of 3125 candidates, totalling 9375 fits
Best hyperparameters (3-fold CV on the training pool):
{'colsample_bytree': 0.7, 'learning_rate': 0.01, 'max_depth': 6, 'n_estimators': 500, '
  reg_lambda': 2.0}
```

Listing 4: Code to set up GridSearchCV with 3-fold CV, optimizing for RMSE.

Using `n_jobs=-1` on my workstation, the 3-fold grid search took about 13 minutes. Once the grid search is done, we then get the parameter combination with the lowest 3-fold cross validation RMSE (on the training set) as output, which is seen in Table 2 below:

Parameter	colsample_bytree	learning_rate	max_depth	reg_lambda	n_estimators
Value	0.7	0.01	6	2	500

Table 2: Parameter combination with the lowest 3-fold CV RMSE (on the training set)

After identifying the optimal hyperparameters using `best_params`, we retrain the XGBoost regressor `XGBRegressor` on the entire training set using these settings, i.e., we refit our model on all of `(X_train_full, y_train_full)`. The final model is then evaluated on the test set, and the resulting RMSE and R^2 scores are reported.

```
xgb_best = xgb.XGBRegressor(
    objective="reg:squarederror",
    random_state=42,
    colsample_bytree=best_params["colsample_bytree"],
    learning_rate=best_params["learning_rate"],
    max_depth=best_params["max_depth"],
    n_estimators=best_params["n_estimators"],
    reg_lambda=best_params["reg_lambda"]
)

xgb_best.fit(X_train_full, y_train_full)
y_pred_best = xgb_best.predict(X_test)

test_rmse_best = np.sqrt(mean_squared_error(y_test, y_pred_best))
test_r2_best = r2_score(y_test, y_pred_best)

print(f"RMSE = {test_rmse_best:.4f}")
print(f"R^2 = {test_r2_best:.4f}")
RMSE = 0.4682
R^2 = 0.4023
```

Listing 5: Code to refit an XGBoost with best hyperparameters on all the training instances.

Here, we see that the tuned XGBoost model achieves a

- Test RMSE: 0.4682
- Test R^2 : 0.4023

The lower RMSE and higher R^2 suggest that the model generalizes better to unseen data after tuning, since both are improved compared to the first model.

Finally, to provide a baseline for evaluating the performance of the XGBoost models, we fit a k -nearest neighbors regressor with $k = 5$ (`KNeighborsRegressor` from `scikit-learn`) on the same training data. The 5-NN model is trained on the full training set and then evaluated on the test set using the same RMSE and R^2 metrics as before. The code is below:

```
knn5 = KNeighborsRegressor(n_neighbors=5)
knn5.fit(X_train_full, y_train_full)
y_pred_knn5 = knn5.predict(X_test)

knn_test_rmse = np.sqrt(mean_squared_error(y_test, y_pred_knn5))
knn_test_r2 = r2_score(y_test, y_pred_knn5)

print(f"RMSE = {knn_test_rmse:.4f}")
print(f"R^2 = {knn_test_r2:.4f}")
RMSE = 0.4798
R^2 = 0.3722
```

Listing 6: Code to fit a k -nearest neighbors regressor with $k = 5$ on the training set.

The results for the 5-NN regressor on the test set are as follows:

- Test RMSE: 0.4798
- Test R^2 : 0.3722

Comparing these metrics to those obtained from the XGBoost models, we observe that the tuned XGBoost regressor achieves a lower RMSE and a higher R^2 score, indicating better predictive performance than the 5-NN baseline for this regression task.

2 A Simple Version of Empirical Bernstein's Inequality

In this exercise, we will derive a bit weaker form of Empirical Bernstein's inequality through a relatively straightforward derivation.

Response to Question 1. We let X and X' be two independent identically distributed random variables. Now, we have to prove that $\mathbb{E}[(X - X')^2] = 2\mathbb{V}[X]$.

Proof. Since X and X' are independent and identically distributed, we first compute

$$\begin{aligned}\mathbb{E}[(X - X')^2] &= \mathbb{E}[X^2 - 2XX' + (X')^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[XX'] + \mathbb{E}[(X')^2].\end{aligned}$$

Because X and X' are independent, then $\mathbb{E}[XX'] = \mathbb{E}[X]\mathbb{E}[X']$. Moreover, since they are

identically distributed, it holds that $\mathbb{E}[X^2] = \mathbb{E}[(X')^2]$ and $\mathbb{E}[X] = \mathbb{E}[X']$. Hence, we get

$$\begin{aligned}\mathbb{E}[(X - X')^2] &= \mathbb{E}[X^2] + \mathbb{E}[X'^2] - 2\mathbb{E}[X]\mathbb{E}[X'] \\ &= 2\mathbb{E}[X^2] - 2(\mathbb{E}[X])^2 \\ &= 2(\mathbb{E}[X^2] - (\mathbb{E}[X])^2) \\ &= 2\mathbb{V}[X],\end{aligned}$$

where we used the definition of the variance, $\mathbb{V}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. $\square$

Response to Question 2. We let $X_1, \dots, X_n$ be independent identically distributed random variables taking values in the $[0, 1]$ interval, and assume that n is even. Also, we let $\hat{\nu}_n = \frac{1}{n} \sum_{i=1}^{n/2} (X_{2i} - X_{2i-1})^2$ and $\nu = \mathbb{V}[X_1]$. We have to prove that

$$\mathbb{P} \left(\nu \geq \hat{\nu}_n + \sqrt{\frac{\ln \frac{1}{\delta}}{n}} \right) \leq \delta.$$

Proof. First, we let $m = \frac{n}{2}$, and for each $i = 1, \dots, m$, we define

$$Y_i = (X_{2i} - X_{2i-1})^2.$$

Because each $X_{2i}, X_{2i-1} \in [0, 1]$, we have that $|X_{2i} - X_{2i-1}| \leq 1$. Hence, $0 \leq Y_i \leq 1$ for $i = 1, \dots, m$. Since $\{X_{2i}, X_{2i-1}\}$ are i.i.d. over all indices, the Y_i 's are themselves independent and identically distributed. In particular, each $Y_i \in [0, 1]$, and $\mathbb{E}[Y_i] = \mathbb{E}[(X_{2i} - X_{2i-1})^2]$. By the same calculation as in Question 1, since X_{2i}, X_{2i-1} are i.i.d. copies of X_1 , we get

$$\mathbb{E}[Y_i] = \mathbb{E}[(X_{2i} - X_{2i-1})^2] = 2\mathbb{V}[X_1] = 2\nu,$$

By definition in the exercise, we have that

$$\hat{\nu}_n = \frac{1}{n} \sum_{i=1}^{n/2} (X_{2i} - X_{2i-1})^2 = \frac{1}{n} \sum_{i=1}^m Y_i.$$

Because $m = n/2 \Leftrightarrow n = 2m$, this can equivalently be written as

$$\hat{\nu}_n = \frac{1}{n} \sum_{i=1}^m Y_i = \frac{1}{2m} \sum_{i=1}^m Y_i.$$

Thus, we have that $\frac{1}{m} \sum_{i=1}^m Y_i = 2\hat{\nu}_n$, and

$$\mathbb{E} \left[\frac{1}{m} \sum_{i=1}^m Y_i \right] = \frac{1}{m} \sum_{i=1}^m \mathbb{E}[Y_i] = \frac{1}{m} m 2\nu = 2\nu.$$

Since each $Y_i \in [0, 1]$ and the Y_i 's are independent, we apply Hoeffding's inequality from Corollary 2.5 in Seldin (2025) to $\sum_{i=1}^m Y_i$. With $\mathbb{E}[Y_i] = 2\nu$, we for every $\varepsilon > 0$ get that

$$\mathbb{P}\left(2\nu - \frac{1}{m} \sum_{i=1}^m Y_i \geq \varepsilon\right) \leq \exp(-2m\varepsilon^2)$$

But since we see that

$$2\nu - \frac{1}{m} \sum_{i=1}^m Y_i = 2\nu - 2\hat{\nu}_n = 2(\nu - \hat{\nu}_n),$$

we can rewrite the bound as

$$\mathbb{P}(2(\nu - \hat{\nu}_n) \geq \varepsilon) = \mathbb{P}\left(\nu - \hat{\nu}_n \geq \frac{\varepsilon}{2}\right) \leq \exp(-2m\varepsilon^2).$$

Since $m = \frac{n}{2}$, we have that $n = 2m$. Hence, $2m\varepsilon^2 = 2\frac{n}{2}\varepsilon^2 = n\varepsilon^2$, such that

$$\mathbb{P}\left(\nu - \hat{\nu}_n \geq \frac{\varepsilon}{2}\right) \leq \exp(-n\varepsilon^2). \quad (\star)$$

We want to show that

$$\nu \geq \hat{\nu}_n + \sqrt{\frac{\ln \frac{1}{\delta}}{n}}$$

with probability at most δ . Equivalently, we see that

$$\nu - \hat{\nu}_n \geq \sqrt{\frac{\ln \frac{1}{\delta}}{n}} \iff \nu - \hat{\nu}_n \geq \frac{\varepsilon}{2} \quad \text{with} \quad \frac{\varepsilon}{2} = \sqrt{\frac{\ln \frac{1}{\delta}}{n}}.$$

That is, we set

$$\varepsilon = 2\sqrt{\frac{\ln \frac{1}{\delta}}{n}}.$$

By plugging this ε into the bound in $(\star)$, it then gives us that

$$\mathbb{P}\left(\nu - \hat{\nu}_n \geq \frac{\varepsilon}{2}\right) = \mathbb{P}\left(\nu - \hat{\nu}_n \geq \sqrt{\frac{\ln \frac{1}{\delta}}{n}}\right) \leq \exp(-n\varepsilon^2).$$

However, as we also observe that

$$n\varepsilon^2 = n \cdot \left(2\sqrt{\frac{\ln \frac{1}{\delta}}{n}}\right)^2 = n \cdot 4 \frac{\ln \frac{1}{\delta}}{n} = 4 \ln \frac{1}{\delta} = \ln \frac{1}{\delta^4},$$

where we used the logarithmic power rule at the last equality sign, it means that

$$\exp(-n\varepsilon^2) = \exp\left(-\ln \frac{1}{\delta^4}\right) = \delta^4.$$

Since $0 < \delta \leq 1$, we know that $\delta^4 \leq \delta$. Therefore, we finally arrive at the desired result by

$$\mathbb{P}\left(\nu - \hat{\nu}_n \geq \sqrt{\frac{\ln \frac{1}{\delta}}{n}}\right) \leq \delta \implies \mathbb{P}\left(\nu \geq \hat{\nu}_n + \sqrt{\frac{\ln \frac{1}{\delta}}{n}}\right) \leq \delta,$$

which concludes the proof. $\square$

Response to Question 3. We let $X_1, \dots, X_n$, n , ν , and $\hat{\nu}_n$ as before, and let $\mu = \mathbb{E}[X_1]$. Now, we have to prove that

$$\mathbb{P}\left(\mu \geq \frac{1}{n} \sum_{i=1}^n X_i + \sqrt{\frac{2\hat{\nu}_n \ln \frac{2}{\delta}}{n}} + \sqrt{2} \left(\frac{\ln \frac{2}{\delta}}{n}\right)^{\frac{3}{4}} + \frac{\ln \frac{2}{\delta}}{3n}\right) \leq \delta.$$

Proof. By Theorem 2.38 in Seldin (2025), Bernstein's Inequality states, for any $\delta \in (0, 1)$,

$$\mathbb{P}\left(\mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \geq \frac{1}{n} \sum_{i=1}^n X_i + \sqrt{\frac{2\nu \ln \frac{1}{\delta}}{n}} + \frac{b \ln \frac{1}{\delta}}{3n}\right) \leq \delta,$$

and here we write $\mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mathbb{E}[X_1] = \mu$ by linearity of expectation and identical distribution. Also, from the theorem, we have that $\mathbb{E}[X_i] - X_i \leq b$ and $\mathbb{V}[X_i] \leq \nu$. Since each $X_i \in [0, 1]$, $b = 1$ works as a deterministic upper bound for $\mathbb{E}[X_i] - X_i \leq 1$, while we from Question 2 know that $\nu = \mathbb{V}[X_i]$. Thus,

$$\mathbb{P}\left(\mu \geq \frac{1}{n} \sum_{i=1}^n X_i + \sqrt{\frac{2\nu \ln \frac{1}{\delta}}{n}} + \frac{\ln \frac{1}{\delta}}{3n}\right) \leq \delta.$$

Now, we define

$$A = \left\{ \mu \geq \frac{1}{n} \sum_{i=1}^n X_i + \sqrt{\frac{2\hat{\nu}_n \ln \frac{2}{\delta}}{n}} + \sqrt{2} \left(\frac{\ln \frac{2}{\delta}}{n}\right)^{3/4} + \frac{\ln \frac{2}{\delta}}{3n} \right\}.$$

We will show that $\mathbb{P}(A) \leq \delta$ by using $\mathbb{P}(A) \leq \mathbb{P}(A \mid B) + \mathbb{P}(\bar{B}) \leq \mathbb{P}(A \mid B) + \frac{\delta}{2}$. From Question 2, we have already shown that, for any $\delta(0, 1)$, it holds that

$$\mathbb{P}\left(\nu \geq \hat{\nu}_n + \sqrt{\frac{\ln \frac{1}{\delta}}{n}}\right) \leq \delta.$$

If we replace δ by $\frac{2}{\delta}$, this becomes

$$\mathbb{P} \left(\nu \geq \hat{\nu}_n + \sqrt{\frac{\ln \frac{2}{\delta}}{n}} \right) \leq \frac{\delta}{2}.$$

We then define the event

$$B = \left\{ \nu \leq \hat{\nu}_n + \sqrt{\frac{\ln \frac{2}{\delta}}{n}} \right\} \Rightarrow \mathbb{P}(\bar{B}) \leq \frac{\delta}{2},$$

where $\bar{B}$ is the complement of B . With B as above and $\varepsilon = \sqrt{\frac{\ln \frac{2}{\delta}}{n}}$, it also means that

$$\sqrt{\frac{2\nu \ln \frac{2}{\delta}}{n}} \leq \sqrt{\frac{2(\hat{\nu}_n + \varepsilon) \ln \frac{2}{\delta}}{n}}.$$

Using the identity $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ for $a, b \geq 0$, and denoting

$$a = \frac{2\hat{\nu}_n \ln \frac{2}{\delta}}{n}, \quad b = \frac{2\varepsilon \ln \frac{2}{\delta}}{n},$$

we obtain

$$\begin{aligned} \sqrt{\frac{2\hat{\nu}_n \ln \frac{2}{\delta}}{n} + \frac{2\varepsilon \ln \frac{2}{\delta}}{n}} &\leq \sqrt{\frac{2\hat{\nu}_n \ln \frac{2}{\delta}}{n}} + \sqrt{\frac{2\varepsilon \ln \frac{2}{\delta}}{n}} \\ &= \sqrt{\frac{2\hat{\nu}_n \ln \frac{2}{\delta}}{n}} + \sqrt{2} \cdot \sqrt{\frac{\varepsilon \ln \frac{2}{\delta}}{n}}. \end{aligned}$$

By inserting ε , it then gives us that

$$\sqrt{\frac{2\hat{\nu}_n \ln \frac{2}{\delta}}{n}} + \sqrt{2} \cdot \sqrt{\frac{\ln \frac{2}{\delta}}{n}} \cdot \sqrt{\frac{\ln \frac{2}{\delta}}{n}} = \sqrt{\frac{2\hat{\nu}_n \ln \frac{2}{\delta}}{n}} + \sqrt{2} \left(\frac{\ln \frac{2}{\delta}}{n} \right)^{3/4}.$$

Therefore, we have that

$$\sqrt{\frac{2\nu \ln \frac{2}{\delta}}{n}} \leq \sqrt{\frac{2\hat{\nu}_n \ln \frac{2}{\delta}}{n}} + \sqrt{2} \left(\frac{\ln \frac{2}{\delta}}{n} \right)^{3/4}.$$

Hence, on the event B , it holds that

$$\left\{ \mu \geq \frac{1}{n} \sum_{i=1}^n X_i + \sqrt{\frac{2\hat{\nu}_n \ln \frac{2}{\delta}}{n}} + \sqrt{2} \left(\frac{\ln \frac{2}{\delta}}{n} \right)^{3/4} + \frac{\ln \frac{2}{\delta}}{3n} \right\}$$

implies

$$\mu \geq \frac{1}{n} \sum_{i=1}^n X_i + \sqrt{\frac{2\nu \ln \frac{2}{\delta}}{n}} + \frac{\ln \frac{2}{\delta}}{3n}.$$

In other words, we have that

$$A|B \subset \left\{ \mu \geq \frac{1}{n} \sum_{i=1}^n X_i + \sqrt{\frac{2\nu \ln \frac{2}{\delta}}{n}} + \frac{\ln \frac{2}{\delta}}{3n} \right\},$$

and by the earlier presented Bernstein's Inequality with $\delta \mapsto \frac{\delta}{2}$, we obtain

$$\mathbb{P} \left(\mu \geq \frac{1}{n} \sum_{i=1}^n X_i + \sqrt{\frac{2\nu \ln \frac{2}{\delta}}{n}} + \frac{\ln \frac{2}{\delta}}{3n} \right) \leq \frac{\delta}{2}.$$

Consequently, we have

$$\mathbb{P}(A | B) \leq \frac{\delta}{2},$$

which then means that

$$\mathbb{P}(A) \leq \mathbb{P}(A | B) + \mathbb{P}(\bar{B}) \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta.$$

Thus, we end up with

$$\mathbb{P} \left(\mu \geq \frac{1}{n} \sum_{i=1}^n X_i + \sqrt{\frac{2\hat{\nu}_n \ln \frac{2}{\delta}}{n}} + \sqrt{2} \left(\frac{\ln \frac{2}{\delta}}{n} \right)^{\frac{3}{4}} + \frac{\ln \frac{2}{\delta}}{3n} \right) \leq \delta,$$

which is the desired result. □

3 PAC-Bayes-Unexpected-Bernstein

Response to Question 1. We let $Z \leq 1$ be a random variable, and have to show that for any $\lambda \in [0, \frac{1}{2}]$, it holds that

$$\mathbb{E} \left[e^{-\lambda Z - \lambda^2 Z^2} \right] \leq e^{-\lambda \mathbb{E}[Z]}.$$

Proof. First, we set $z = -\lambda Z$. Since $0 \leq \lambda \leq \frac{1}{2}$ and $0 \leq Z \leq 1$, we have that

$$\lambda Z \leq \frac{1}{2} \implies z = -\lambda Z \geq -\frac{1}{2},$$

so we may apply the inequality

$$z - z^2 \leq \ln(1 + z),$$

which is valid for all $z \geq -\frac{1}{2}$. Exponentiating both sides gives

$$e^{z-z^2} \leq 1+z \implies e^{-\lambda Z - \lambda^2 Z^2} \leq 1 - \lambda Z.$$

Taking expectations, then gives us that

$$\mathbb{E} \left[e^{-\lambda Z - \lambda^2 Z^2} \right] \leq \mathbb{E}[1 - \lambda Z] = 1 - \lambda \mathbb{E}[Z].$$

Here, we needed $\lambda \in [0, \frac{1}{2}]$ and $Z \leq 1$ to ensure $-\lambda Z \geq -\frac{1}{2}$, so that $z - z^2 \leq \ln(1+z)$ applies. We used $Z \leq 1$ again when bounding $\lambda Z \leq \frac{1}{2}$. Finally, for any real u , we have the bound $1+u \leq e^u$. By applying this with $u = -\lambda \mathbb{E}[Z]$. Then, we get that

$$1 - \lambda \mathbb{E}[Z] = 1 + (-\lambda \mathbb{E}[Z]) \leq e^{-\lambda \mathbb{E}[Z]}.$$

By then combining the two steps, it yields

$$\mathbb{E} \left[e^{-\lambda Z - \lambda^2 Z^2} \right] \leq 1 - \lambda \mathbb{E}[Z] \leq e^{-\lambda \mathbb{E}[Z]},$$

as claimed. □

Response to Question 2. For $Z \leq 1$ and $\lambda \in [0, \frac{1}{2}]$, we have to prove that

$$\mathbb{E} \left[e^{\lambda(\mathbb{E}[Z] - Z) - \lambda^2 Z^2} \right] \leq 1.$$

Proof. In Question 1, we showed that

$$\mathbb{E} \left[e^{-\lambda Z - \lambda^2 Z^2} \right] \leq e^{-\lambda \mathbb{E}[Z]}$$

for $Z \leq 1$ and $\lambda \in [0, \frac{1}{2}]$. But since $\mathbb{E}[Z]$ is just a constant, we can write

$$\begin{aligned} \mathbb{E} \left[e^{\lambda(\mathbb{E}[Z] - Z) - \lambda^2 Z^2} \right] &= \mathbb{E} \left[e^{\lambda \mathbb{E}[Z] - \lambda Z - \lambda^2 Z^2} \right] \\ &= e^{\lambda \mathbb{E}[Z]} \mathbb{E} \left[e^{-\lambda Z - \lambda^2 Z^2} \right] \\ &\leq e^{\lambda \mathbb{E}[Z]} e^{-\lambda \mathbb{E}[Z]} \\ &= e^{\lambda \mathbb{E}[Z] - \lambda \mathbb{E}[Z]} \\ &= e^0 \\ &= 1, \end{aligned}$$

which is exactly the desired inequality. □

Response to Question 3. We let $Z_1, \dots, Z_n$ be independent random variables upper bounded by 1. For any $\lambda \in [0, \frac{1}{2}]$, we have to show that

$$\mathbb{E} \left[e^{\lambda \sum_{i=1}^n (\mathbb{E}[Z_i] - Z_i) - \lambda^2 \sum_{i=1}^n Z_i^2} \right] \leq 1.$$

Proof. First, we observe that

$$\exp \left(\lambda \sum_{i=1}^n (\mathbb{E}[Z_i] - Z_i) - \lambda^2 \sum_{i=1}^n Z_i^2 \right) = \exp \left(\sum_{i=1}^n [\lambda (\mathbb{E}[Z_i] - Z_i) - \lambda^2 Z_i^2] \right).$$

Since $\exp(\sum_{i=1}^n a_i) = \prod_{i=1}^n e^{a_i}$, choosing $a_i = \lambda (\mathbb{E}[Z_i] - Z_i) - \lambda^2 Z_i^2$ yields

$$\exp \left(\sum_{i=1}^n [\lambda (\mathbb{E}[Z_i] - Z_i) - \lambda^2 Z_i^2] \right) = \prod_{i=1}^n \exp (\lambda (\mathbb{E}[Z_i] - Z_i) - \lambda^2 Z_i^2)$$

Since $Z_1, \dots, Z_n$ are independent random variables, we thus factor the joint moment by

$$\mathbb{E} \left[\prod_{i=1}^n \exp (\lambda (\mathbb{E}[Z_i] - Z_i) - \lambda^2 Z_i^2) \right] = \prod_{i=1}^n \mathbb{E} \left[e^{\lambda (\mathbb{E}[Z_i] - Z_i) - \lambda^2 Z_i^2} \right]$$

From Question 2 (with each $Z_i \leq 1$ and $\lambda \in [0, \frac{1}{2}]$), we for every $i = 1, \dots, n$ have that

$$\mathbb{E} \left[e^{\lambda (\mathbb{E}[Z_i] - Z_i) - \lambda^2 Z_i^2} \right] \leq 1.$$

Hence, by applying this bound, we obtain

$$\prod_{i=1}^n \underbrace{\mathbb{E} \left[e^{\lambda (\mathbb{E}[Z_i] - Z_i) - \lambda^2 Z_i^2} \right]}_{\leq 1} \leq 1^n = 1,$$

which is exactly the desired inequality

$$\prod_{i=1}^n \mathbb{E} \left[e^{\lambda (\mathbb{E}[Z_i] - Z_i) - \lambda^2 Z_i^2} \right] = \mathbb{E} \left[e^{\lambda \sum_{i=1}^n (\mathbb{E}[Z_i] - Z_i) - \lambda^2 \sum_{i=1}^n Z_i^2} \right] \leq 1.$$

□

Response to Question 4. We let $Z_1, \dots, Z_n$ be independent random variables upper bounded by 1. For any $\lambda \in (0, \frac{1}{2}]$, we have to show that

$$\mathbb{P} \left(\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n Z_i \right] \geq \frac{1}{n} \sum_{i=1}^n Z_i + \frac{\lambda}{n} \sum_{i=1}^n Z_i^2 + \frac{\ln \frac{1}{\delta}}{\lambda n} \right) \leq \delta.$$

Proof. First, we write

$$S = \sum_{i=1}^n Z_i, \quad T = \sum_{i=1}^n Z_i^2,$$

while we let $\mu = S/n$ and $\bar{\mu} = \mathbb{E}[S]/n$. For any $\lambda \in (0, \frac{1}{2}]$ and $\delta > 0$, we wish to show that

$$\mathbb{P} \left(\bar{\mu} \geq \mu + \frac{\lambda}{n} T + \frac{1}{\lambda n} \ln \frac{1}{\delta} \right) \leq \delta.$$

Equivalently, that the event

$$\bar{\mu} - \mu - \frac{\lambda}{n}T \geq \frac{1}{\lambda n} \ln \frac{1}{\delta}$$

has probability at most δ . We multiply both sides by $\lambda n > 0$ to get the same event as

$$\lambda(\mathbb{E}[S] - S) - \lambda^2 T \geq \ln \frac{1}{\delta}.$$

Hence, on this event, we have that

$$\exp(\lambda(\mathbb{E}[S] - S) - \lambda^2 T) \geq \exp\left(\ln \frac{1}{\delta}\right) = \frac{1}{\delta}$$

By Markov's inequality from Theorem 2.1 in Seldin (2025), we then get that

$$\mathbb{P}\left(X \geq \ln \frac{1}{\delta}\right) = \mathbb{P}\left(e^X \geq \frac{1}{\delta}\right) \leq \delta \mathbb{E}[e^X]$$

with $X = \lambda(\mathbb{E}[S] - S) - \lambda^2 T$. But from Question 3, we know that

$$\mathbb{E}[e^X] = \mathbb{E}\left[e^{\lambda \sum_{i=1}^n (\mathbb{E}[Z_i] - Z_i) - \lambda^2 \sum_{i=1}^n Z_i^2}\right] \leq 1,$$

so by putting these together, it then gives us that

$$\mathbb{P}\left(\bar{\mu} \geq \mu + \frac{\lambda}{n}T + \frac{1}{\lambda n} \ln \frac{1}{\delta}\right) = \mathbb{P}\left(X \geq \ln \frac{1}{\delta}\right) \leq \delta \mathbb{E}[e^X] \leq \delta \cdot 1 = \delta$$

as claimed. □

Response to Question 5. We let $\Lambda = \{\lambda_1, \dots, \lambda_k\}$ be a grid of k values of λ , such that $\lambda_i \in (0, \frac{1}{2}]$ for all i . We have to prove the unexpected Bernstein inequality given by

$$\mathbb{P}\left(\mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n Z_i\right] \geq \frac{1}{n} \sum_{i=1}^n Z_i + \min_{\lambda \in \Lambda} \left(\frac{\lambda}{n} \sum_{i=1}^n Z_i^2 + \frac{\ln \frac{k}{\delta}}{\lambda n}\right)\right) \leq \delta.$$

Proof. First, we fix a grid $\Lambda = \{\lambda_1, \dots, \lambda_k\} \subset (0, \frac{1}{2}]$. Then, define the event

$$A = \left\{ \bar{Z} \geq Z + \min_{\lambda \in \Lambda} \left(\frac{\lambda}{n} T + \frac{1}{\lambda n} \ln \frac{k}{\delta} \right) \right\},$$

where we write $\bar{Z} = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n Z_i\right]$, $Z = \sum_{i=1}^n Z_i$ and $T = \sum_{i=1}^n Z_i^2$. By definition,

$$A = \left\{ \exists \lambda \in \Lambda: \bar{Z} \geq Z + \frac{\lambda}{n} T + \frac{1}{\lambda n} \ln \frac{k}{\delta} \right\}.$$

For each fixed $\lambda \in \Lambda$, we let

$$A_\lambda = \left\{ \bar{Z} \geq Z + \frac{\lambda}{n}T + \frac{1}{\lambda n} \ln \frac{k}{\delta} \right\},$$

and then we observe that

$$A = \bigcup_{\lambda \in \Lambda} A_\lambda.$$

By the result of Question 4 (with δ replaced by $\frac{\delta}{k}$), for each fixed $\lambda \in \Lambda$, we have that

$$\mathbb{P}(A_\lambda) = \mathbb{P}\left(\bar{Z} \geq Z + \frac{\lambda}{n}T + \frac{1}{\lambda n} \ln \frac{k}{\delta}\right) \leq \frac{\delta}{k}$$

Applying the union bound over $\lambda \in \Lambda$ then gives us that

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{\lambda \in \Lambda} A_\lambda\right) \leq \sum_{\lambda \in \Lambda} \mathbb{P}(A_\lambda) \leq k \cdot \frac{\delta}{k} = \delta,$$

since there exactly k different λ 's in our grid Λ , and for each one we showed $\mathbb{P}(A) \leq \frac{\delta}{k}$, so when we sum up those k identical upper bounds, we get $\sum_{\lambda \in \Lambda} \mathbb{P}(A_\lambda) \leq \sum_{\lambda \in \Lambda} \frac{\delta}{k} = k \frac{\delta}{k} = \delta$.

This is exactly the claimed

$$\mathbb{P}\left(\bar{Z} \geq Z + \min_{\lambda \in \Lambda} \left(\frac{\lambda}{n}T + \frac{\ln \frac{k}{\delta}}{\lambda n}\right)\right) \leq \delta.$$

□

Response to Question 6. We will perform an empirical comparison of the kl and Unexpected Bernstein inequalities. First, we take $n = 100$ i.i.d. samples $Z_1, \dots, Z_{100}$ of ternary random variables that take three values in the set $\{0, \frac{1}{2}, 1\}$. We let the probabilities

$$p_0 = \mathbb{P}(Z = 0), \quad p_{\frac{1}{2}} = \mathbb{P}(Z = \frac{1}{2}), \quad \text{and} \quad p_1 = \mathbb{P}(Z = 1),$$

where $p_0 + p_{\frac{1}{2}} + p_1 = 1$. Hereafter, we set

$$p_0 = p_1 = \frac{1 - p_{\frac{1}{2}}}{2},$$

such that the probabilities of $Z = 0$ and $Z = 1$ are equal, with one parameter $p_{\frac{1}{2}}$ that controls the probability mass of the central value. By letting $p = \mathbb{E}[Z]$, we see that

$$p = \mathbb{E}[Z] = 0 \cdot p_0 + \frac{1}{2} \cdot p_{\frac{1}{2}} + 1 \cdot p_1 = \frac{1}{2},$$

because $p_0 = p_1$ in this constructed setup. We will compare the bounds as a function of $p_{\frac{1}{2}} \in [0, 1]$. For each value of $p_{\frac{1}{2}}$ in a grid covering the $[0, 1]$ interval, we draw a random sample $Z_1, \dots, Z_{100}$ from our constructed distribution, and then we let $\hat{p}_{100} = \frac{1}{100} \sum_{i=1}^{100} X_i$ and $\hat{v}_{100} = \frac{1}{100} \sum_{i=1}^{100} Z_i^2$. We now present the two bounds:

1. **The kl bound:** By Theorem 2.27 with the kl inequality, for any $\delta \in (0, 1)$ and $\varepsilon = \frac{\ln \frac{1}{\delta}}{n}$, with probability at least $1 - \delta$, it holds that $\text{kl}(\hat{p}_n || p) \leq \varepsilon$. Since $p = \frac{1}{2}$ is known, we invert $p \mapsto \text{kl}(\hat{p}_{100} || p)$ on $[\hat{p}_{100}, 1]$. We define the upper inverse of kl by

$$\text{kl}^{-1+}(\hat{p}_{100}, \varepsilon) := \max\{p: p \in [\hat{p}_{100}, 1]: \text{kl}(\hat{p}_{100} || p) \leq \varepsilon\}.$$

Then, we get the kl bound on $p - \hat{p}_n$ by

$$p \leq \text{kl}^{-1+}(\hat{p}_{100}, \varepsilon) \Rightarrow p - \hat{p}_{100} \leq \text{kl}^{-1+}(\hat{p}_{100}, \varepsilon) - \hat{p}_{100}.$$

2. **Unexpected-Bernstein bound:** Recalling from Question 5, we pick a finite grid $\Lambda = \{\frac{1}{2}, \frac{1}{2^2}, \dots, \frac{1}{2^k}\}$ with $|\Lambda| = k = \lceil \log_2(\sqrt{n/\ln(1/\delta)})/2 \rceil$, where we have $n = 100$ and choose $\delta = 0.05$. Then, the Unexpected Bernstein bound on $p - \hat{p}_{100}$ is given by

$$\min_{\lambda \in \Lambda} \left(\lambda \hat{v}_n + \frac{\ln \frac{k}{\delta}}{\lambda n} \right) = \min_{\lambda \in \Lambda} \left(\lambda \hat{v}_{100} + \frac{\ln \frac{\lceil \log_2(\sqrt{100/\ln(1/0.05)})/2 \rceil}{0.05}}{\lambda \cdot 100} \right).$$

There is no closed-form expression to compute $\text{kl}^{-1+}(\hat{p}_{100}, \varepsilon)$, so we use a binary-search algorithm to find $p = \text{kl}(\hat{p}_{100}, \varepsilon) = \max\{p: p \in [\hat{p}_{100}, 1] \text{ and } \text{kl}(\hat{p}_{100} || p) \leq \varepsilon\}$ numerically:

Algorithm 1 Compute the upper inverse $\text{kl}^{-1+}(\hat{p}_n, \varepsilon)$ by binary search

Require: Empirical average $\hat{p}_n \in [0, 1]$, tolerance $\varepsilon > 0$, search precision $\tau > 0$

Ensure: Approximate solution $p \approx \max\{p: p \in [\hat{p}_n, 1] \text{ and } \text{kl}(\hat{p}_n || p) \leq \varepsilon\}$

```

1:  $a \leftarrow \hat{p}_n$                                 ▷ initialize lower (feasible) bound
2:  $b \leftarrow 1$                                 ▷ initialize upper (infeasible) bound
3: while  $b - a > \tau$  do                          ▷ iterate until desired precision
4:    $m \leftarrow \frac{a+b}{2}$ 
5:    $D \leftarrow \text{kl}(\hat{p}_n || m)$                     ▷ compute kl-divergence
6:   if  $D > \varepsilon$  then                               ▷ midpoint violates bound → too large
7:      $b \leftarrow m$ 
8:   else                                           ▷ midpoint is valid → still feasible
9:      $a \leftarrow m$ 
10:  end if
11: end while
12: return  $a$                                      ▷ final approximation within tolerance
```

We then generate a figure (the code can be found in `question_3_10_6.py`), where we both plot the Unexpected Bernstein bound on $p - \hat{p}_{100}$ and the kl bound on $p - \hat{p}_{100}$ as a function of $p_{\frac{1}{2}}$ for $p_{\frac{1}{2}} \in [0, 1]$, which can be seen in Figure 2 below:

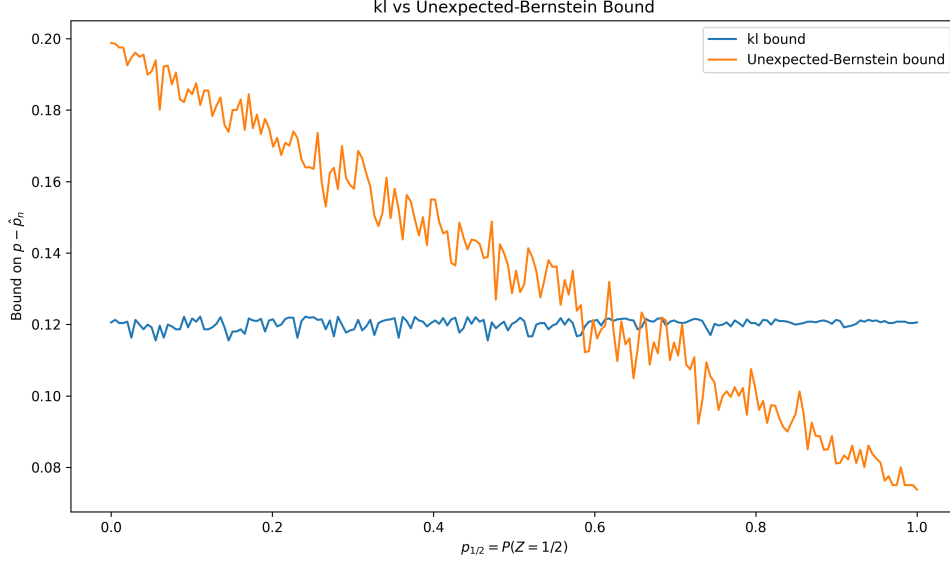


Figure 2: Comparison of bounds on $p - \hat{p}_n$ as a function of $\hat{p}_{\frac{1}{2}} \in [0, 1]$

As seen, the kl bound remains essentially constant at about 0.12 for all $p_{\frac{1}{2}}$, whereas the Unexpected-Bernstein bound starts near 0.20 at $p_{\frac{1}{2}} = 0$ and then decreases as $\hat{v}_{100}$ shrinks, reaching approx. 0.075 at $p_{\frac{1}{2}} = 1$. Consequently, the Unexpected-Bernstein bound is looser than the kl bound when $p_{\frac{1}{2}}$ is around 0.6, but becomes strictly tighter for larger $p_{\frac{1}{2}}$.

Response to Question 7. We let S be an i.i.d. sample, h a prediction rule, and $\ell(y', y)$ a loss function upper bounded by 1. We then define $\hat{V}(h, S) = \frac{1}{n} \sum_{i=1}^n \ell(h(X_i), Y_i)^2$, and also $L(h)$ and $\hat{L}(h, S)$ as usual. For any $\lambda \in [0, \frac{1}{2}]$, we have to show that

$$\mathbb{E} \left[e^{n(\lambda(L(h) - \hat{L}(h, S)) - \lambda^2 \hat{V}(h, S))} \right] \leq 1.$$

Proof. In Question 3, we showed that if $Z_1, \dots, Z_n$ are independent random variables upper bounded by 1, then for any $\lambda \in [0, \frac{1}{2}]$, we have that

$$\mathbb{E} \left[e^{\lambda \sum_{i=1}^n (\mathbb{E}[Z_i] - Z_i) - \lambda^2 \sum_{i=1}^n Z_i^2} \right] \leq 1.$$

Now, we set $Z_i = \ell(h(X_i), Y_i)$, so that $0 \leq Z_i \leq 1$, since $\ell \leq 1$, and by the definition of the true risk, we have $\mathbb{E}[Z_i] = \mathbb{E}[\ell(h(X_i), Y_i)] = L(h)$, while

$$\frac{1}{n} \sum_{i=1}^n Z_i = \frac{1}{n} \sum_{i=1}^n \ell(h(X_i), Y_i) = \hat{L}(h, S), \quad \frac{1}{n} \sum_{i=1}^n Z_i^2 = \frac{1}{n} \sum_{i=1}^n \ell(h(X_i), Y_i)^2 = \hat{V}(h, S).$$

Then, we see that

$$\sum_{i=1}^n (\mathbb{E}[Z_i] - Z_i) = \sum_{i=1}^n (L(h) - Z_i) = nL(h) - \sum_{i=1}^n Z_i = n(L(h) - \hat{L}(h, S)),$$

and

$$\sum_{i=1}^n Z_i^2 = n\hat{V}(h, S).$$

Substituting these two expressions into the bound from Question 3, gives, for any $\lambda \in [0, \frac{1}{2}]$,

$$\mathbb{E} \left[e^{\lambda \sum_{i=1}^n (\mathbb{E}[Z_i] - Z_i) - \lambda^2 \sum_{i=1}^n Z_i^2} \right] = \mathbb{E} \left[e^{n(\lambda(L(h) - \hat{L}(h, S)) - \lambda^2 \hat{V}(h, S))} \right] \leq 1,$$

which is exactly the desired bound. $\square$

Response to Question 8. We let S and ℓ be as before. Now, we let $\mathcal{H}$ be a set of prediction rules, and π be a distribution on $\mathcal{H}$ that is independent of S .

For any $\lambda \in (0, \frac{1}{2}]$, we have to show that

$$\mathbb{P} \left(\exists \rho: \mathbb{E}_\rho[L(h)] \geq \mathbb{E}_\rho[\hat{L}(h, S)] + \lambda \mathbb{E}_\rho[\hat{V}(h, S)] + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{1}{\delta}}{n\lambda} \right) \leq \delta,$$

where ρ denotes a distribution on $\mathcal{H}$.

Proof. First, we fix any prior distribution π over $\mathcal{H}$ (independent of the sample S) and any $\lambda \in (0, \frac{1}{2}]$, and then we define the function

$$f(h, S) = n \left(\lambda(L(h) - \hat{L}(h, S)) - \lambda^2 \hat{V}(h, S) \right).$$

From Question 7, we know that for each fixed h , it holds that

$$\mathbb{E}_S [e^{f(h, S)}] \leq 1.$$

Since h is drawn from π independently of S , then the Fubini–Tonelli theorem lets us interchange the order of integration, such that

$$\mathbb{E}_S [\mathbb{E}_{h \sim \pi} [e^{f(h, S)}]] = \mathbb{E}_{h \sim \pi} [\mathbb{E}_S [e^{f(h, S)}]] \leq 1.$$

In particular, with probability 1 over S , we have

$$\mathbb{E}_{h \sim \pi} [e^{f(h, S)}] \leq 1.$$

Next, we invoke the PAC-Bayes lemma from Lemma 3.28 in Seldin (2025) with our choice of f and π . Since $\mathbb{E}_{h \sim \pi} [e^{f(h, S)}] \leq 1$ deterministically, it tells us that for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the draw of S , for every posterior ρ on $\mathcal{H}$ simultaneously,

$$\mathbb{E}_{h \sim \rho} [f(h, S)] \leq \text{KL}(\rho \parallel \pi) + \ln \frac{1}{\delta}.$$

But by the definition of f , we have that

$$\mathbb{E}_{h \sim \rho}[f(h, S)] = n \left[\lambda \left(\mathbb{E}_\rho[L(h)] - \mathbb{E}_\rho[\hat{L}(h, S)] \right) - \lambda^2 \mathbb{E}_\rho[\hat{V}(h, S)] \right],$$

such that

$$n \left[\lambda \left(\mathbb{E}_\rho[L(h)] - \mathbb{E}_\rho[\hat{L}(h, S)] \right) - \lambda^2 \mathbb{E}_\rho[\hat{V}(h, S)] \right] \leq \text{KL}(\rho \parallel \pi) + \ln \frac{1}{\delta}.$$

By dividing by $n\lambda > 0$ and moving $\mathbb{E}_\rho[\hat{L}(h, S)]$ to the right, it then gives, for every ρ ,

$$\mathbb{E}_\rho[L(h)] \leq \mathbb{E}_\rho[\hat{L}(h, S)] + \lambda \mathbb{E}_\rho[\hat{V}(h, S)] + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{1}{\delta}}{n\lambda}.$$

Since this holds for all ρ simultaneously with probability at least $1 - \delta$, we conclude that

$$\mathbb{P} \left(\exists \rho: \mathbb{E}_\rho[L(h)] \geq \mathbb{E}_\rho[\hat{L}(h, S)] + \lambda \mathbb{E}_\rho[\hat{V}(h, S)] + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{1}{\delta}}{n\lambda} \right) \leq \delta,$$

which is exactly the desired result. $\square$

Response to Question 9. We let $\Lambda = \{\lambda_1, \dots, \lambda_k\}$ be a grid of k values of λ , such that $\lambda_i \in (0, \frac{1}{2}]$ for all i . Also, we let S , ℓ and π be as before. Now, we have to prove that

$$\mathbb{P} \left(\exists \rho: \mathbb{E}_\rho[L(h)] \geq \mathbb{E}_\rho[\hat{L}(h, S)] + \min_{\lambda \in \Lambda} \left(\lambda \mathbb{E}_\rho[\hat{V}(h, S)] + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{k}{\delta}}{n\lambda} \right) \right) \leq \delta.$$

Proof. Let $\Lambda = \{\lambda_1, \dots, \lambda_k\} \subset (0, \frac{1}{2}]$. By Question 8, for any fixed $\lambda \in (0, \frac{1}{2}]$ and prior π , with probability at least $1 - \delta'$ over S , one has simultaneously for all posteriors ρ ,

$$\mathbb{E}_\rho[L(h)] \leq \mathbb{E}_\rho[\hat{L}(h, S)] + \lambda \mathbb{E}_\rho[\hat{V}(h, S)] + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{1}{\delta}}{n\lambda}.$$

We apply this with $\delta' = \delta/k$ at each $\lambda_i \in \Lambda$. For each $i = 1, \dots, k$, we then define the event

$$E_i := \left\{ \forall \rho: \mathbb{E}_\rho[L(h)] \leq \mathbb{E}_\rho[\hat{L}(h, S)] + \lambda_i \mathbb{E}_\rho[\hat{V}(h, S)] + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{k}{\delta}}{n\lambda_i} \right\},$$

where

$$\mathbb{P}(E_i) \geq 1 - \frac{\delta}{k}.$$

Hence, by the union bound, we get that

$$\mathbb{P} \left(\bigcap_{i=1}^k E_i \right) = 1 - \mathbb{P} \left(\bigcup_{i=1}^k E_i^c \right) \geq 1 - \sum_{i=1}^k \frac{\delta}{k} = k \frac{\delta}{k} = 1 - \delta.$$

Thus, with probability at least $1 - \delta$, for every ρ and every $\lambda_i \in \Lambda$,

$$\mathbb{E}_\rho[L(h)] \leq \mathbb{E}_\rho[\hat{L}(h, S)] + \lambda_i \mathbb{E}_\rho[\hat{V}(h, S)] + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{k}{\delta}}{n \lambda_i}.$$

But then for each ρ , we have that

$$\mathbb{E}_\rho[L(h)] \leq \mathbb{E}_\rho[\hat{L}(h, S)] + \min_{\lambda \in \Lambda} \left(\lambda \mathbb{E}_\rho[\hat{V}(h, S)] + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{k}{\delta}}{n \lambda} \right).$$

Since $\mathbb{P}(\bigcap_{i=1}^k E_i) \geq 1 - \delta$, then $\mathbb{P}((\bigcap_{i=1}^k E_i)^c) \leq \delta$, but $(\bigcap_{i=1}^k E_i)^c$ is exactly that there exists some posterior ρ for which none of the k inequalities holds, which is equivalent to

$$\exists \rho: \quad \mathbb{E}_\rho[L(h)] \geq \mathbb{E}_\rho[\hat{L}(h, S)] + \min_{\lambda \in \Lambda} \left(\lambda \mathbb{E}_\rho[\hat{V}(h, S)] + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{k}{\delta}}{n \lambda} \right),$$

so by taking complements, it gives us

$$\mathbb{P} \left(\exists \rho: \mathbb{E}_\rho[L(h)] \geq \mathbb{E}_\rho[\hat{L}(h, S)] + \min_{\lambda \in \Lambda} \left(\lambda \mathbb{E}_\rho[\hat{V}(h, S)] + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{k}{\delta}}{n \lambda} \right) \right) \leq \delta,$$

which is the desired PAC–Bayes–Unexpected–Bernstein inequality. $\square$

References

Seldin, Y. (2025). Machine Learning: The Science of Selection under Uncertainty.